Izvod

$$V_a(x) = \begin{cases} +\infty & ; |x-\frac{L}{2}| \geq \frac{L}{2} \\ 0 & ; |x-\frac{L}{2}| < \frac{L}{2} \end{cases} \qquad \text{simetrija} \quad V_S(x) = \frac{V_0}{L}x$$

probna funk: $\Psi(x,L) = \begin{cases} Ax(x-L)e^{-Lx} & ; \ 0<x<L \\ 0 \end{cases}$

lokalna energija $E_L(x,L) = \frac{\hat{H}\Psi(x,L)}{\Psi(x,L)}$ $\quad \hat{H} = -\frac{\hbar^2}{2M}\frac{d^2}{dx^2} + \frac{V_0}{L}x$

$$E_L(x,L) = -\frac{\hbar^2}{2M}\frac{\Psi''(x,L)}{\Psi(x,L)} + \frac{V_0}{L}x$$

$$\Psi(x,L) = Ax(x-L)e^{-Lx}$$

$$x(x-L) = x^2 - Lx$$

$$\Psi(x,L) = A(x^2-Lx)e^{-Lx}$$

1° der: $f(x) = x^2 - Lx$, $g(x) = e^{-Lx}$

$$f'(x) = 2x - L$$

$$g'(x) = -Le^{-Lx}$$

$$\Psi'(x,L) = A\frac{d}{dx}\left[(x^2-Lx)e^{-Lx}\right]$$

$$\Psi'(x,L) = A\left[(2x-L)e^{-Lx} + (x^2-Lx)(-L)e^{-Lx}\right]$$

$$\Psi'(x,L) = Ae^{-Lx}\left[2x - L - Lx(x-L)\right]$$

2° der: $h(x) = 2x - L - L(x^2-Lx)$

$$h'(x) = \frac{d}{dx}\left[2x - L - L(x^2-Lx)\right]$$

$$h'(x) = 2 - L(2x-L)$$

$$\Psi'(x,L) = A\frac{d}{dx}\left[e^{-Lx}h(x)\right]$$

$$\Psi''(x,L) = Ae^{-Lx}\left[h'(x) - Lh(x)\right]$$

$$\Psi''(x,L) = Ae^{-Lx}\left[2 - 4Lx + 2L^2 + L^2(x^2-Lx)\right]$$

$$\frac{\Psi''(x,L)}{\Psi(x,L)} = \frac{Ae^{-Lx}\left[2 - 4Lx + 2L^2 + L^2x(x-L)\right]}{Ax(x-L)e^{-Lx}} = \frac{2 - 4Lx + 2L^2 + L^2x(x-L)}{x(x-L)}$$

$$= \frac{2 - 4Lx + 2L^2}{x(x-L)} + \frac{L^2x(x-L)}{x(x-L)}$$

$$= L^2 + 2\frac{1 + L^2 - 2Lx}{x(x-L)}$$

$$E_L(x,\alpha) = -\frac{\hbar^2}{2m}\left[\alpha^2 + \frac{2+2\alpha L - 4\alpha x}{x(x-L)}\right] + \frac{V_0}{L}x$$

$$= -\frac{\hbar^2\alpha^2}{2m} - \frac{\hbar^2}{m}\,\frac{1+\alpha L - 2\alpha x}{x(x-L)} + \frac{V_0}{L}x$$

qbo $\hbar = 1$

$m = 1$

$\frac{\hbar^2}{2m} = \frac{1}{2}$

$$E_L(x,\alpha) = -\frac{1}{2}\alpha^2 - \frac{1+\alpha L - 2\alpha x}{x(x-L)} + \frac{V_0}{L}x$$